TGFB1 promoter-reporter architectures

5' -> 3' display, genomic 41358961 -> 41207203 | strand - | assembly GRCh38 | annotation Ensembl live GRCh38 | upstream 5000 bp | downstream 1000 bp

Canonical reporter comparison: 1 TSS class(es), 3 transcript(s), 9 explicit architecture row(s); normalized evidence remains observational/predictive context.

Legend (annotation-derived transcript features)

TSS stretches: reference for similarity below

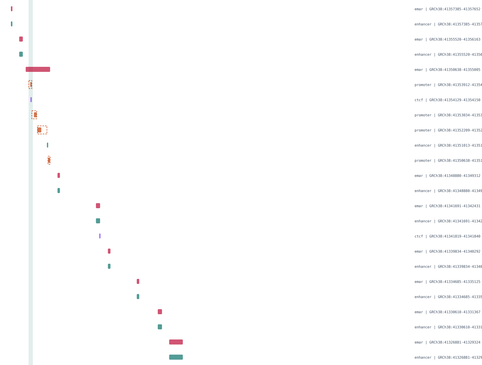
TGFB1_tss_stretch_1 — 41,353,722..41,354,461

Transcript models and annotation-derived metrics

Proposed reporter architectures

Ensembl-annotated regulatory regions

Ensembl Regulation | release 2026-08 | pipeline 2.1 | GRCh38



26 additional source-bound row(s) omitted from SVG; complete rows remain in JSON.

Projected occupancy grouped by declared biological context

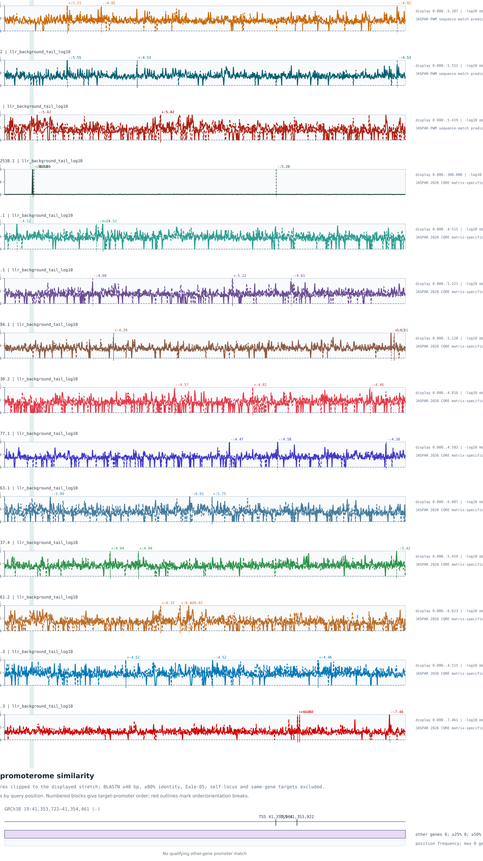
TP73 occupancy — SAOS-2 | shared within group | observed group max 25.000

TP73 occupancy — SK-MEL-29-2 | shared within group | observed group max 13.000

H3K4me3 chromatin context — SAOS-2 | shared within group | observed group max 31.000

H3K4me3 chromatin context — SK-MEL-29-2 | shared within group | observed group max 32.000

Predicted TF binding scores



TSS-local Ensembl-feature promoterome similarity

Transcript-oriented −500/+200 bp; features clipped to the displayed stretch; BLASTN w40 bp, ≥80% identity, ExSe-05; self-locus and same-gene targets excluded.

Blue frequency strips count distinct other genes by query position. Numbered blocks give target-promoter order; red outlines mark order/orientation breaks.

TGFB1_tss_stretch_1

Similarity is structural evidence: Ensembl classes, predicted TFBS and CUT&RUN enrichment do not prove reporter activity or sufficiency.

Reporter interpretation boundaries

Transcript annotation alone does not establish which TSS is used in the assayed cells.
CUT&RUN overlap is occupancy/enrichment evidence, not proof of TSS usage, promoter activity, direct regulation, biochemical affinity, or luciferase output.
Reporter geometry does not verify the identity of a physical laboratory construct.

Warnings / interpretation checks

Evidence provenance

ensembl_regulation_annotation_index: ensembl_regulation_2026_08_grch38
ensembl_regulation_annotation_intervals: ensembl_regulation_2026_08_grch38
isoform_panel: TGFB1_panel
occupancy_layout: tp73_saos2_tp73_skmel29_2_h3k4me3_saos2_h3k4me3_skmel29_2
predicted_tf_score_jaspar_pwm: local_jaspar_registry
projected_occupancy_track: E-MTAB-15709-h3k4me3-saos2-DN-01
projected_occupancy_track: E-MTAB-15709-h3k4me3-saos2-GFP-01
projected_occupancy_track: E-MTAB-15709-h3k4me3-saos2-TA-01
projected_occupancy_track: E-MTAB-15709-h3k4me3-skmel29-2-DN-01
projected_occupancy_track: E-MTAB-15709-h3k4me3-skmel29-2-GFP-01